

Leveraging Allophony in Self-Supervised Speech Models for Atypical Pronunciation Assessment



Kwanghee Choi¹ kwanghee@cmu.edu Eunjung Yeo¹ Calvin Chang¹ Shinji Watanabe¹ David Mortensen

¹Language Technology Institute at Carnegie Mellon University

Background

What is allophony?

- A **phoneme** is a unit of sound that can make meaningful distinctions in a specific language.
- Example. We distinguish /pæt/ (*pat*) and /kæt/ (*cat*) because /p/ and /k/ is a different phoneme.
- Allophones** are contextual realizations of phonemes.
- Example. English phoneme /t/ is realized as multiple allophones: [t^h] (aspirated stop) in *tap*, [t] (unaspirated stop) in *stop*, [ɾ] (flap) in *butter*, and [ɤ] (glottal stop) in *kitten*.

What are self-supervised speech models (S3Ms)?

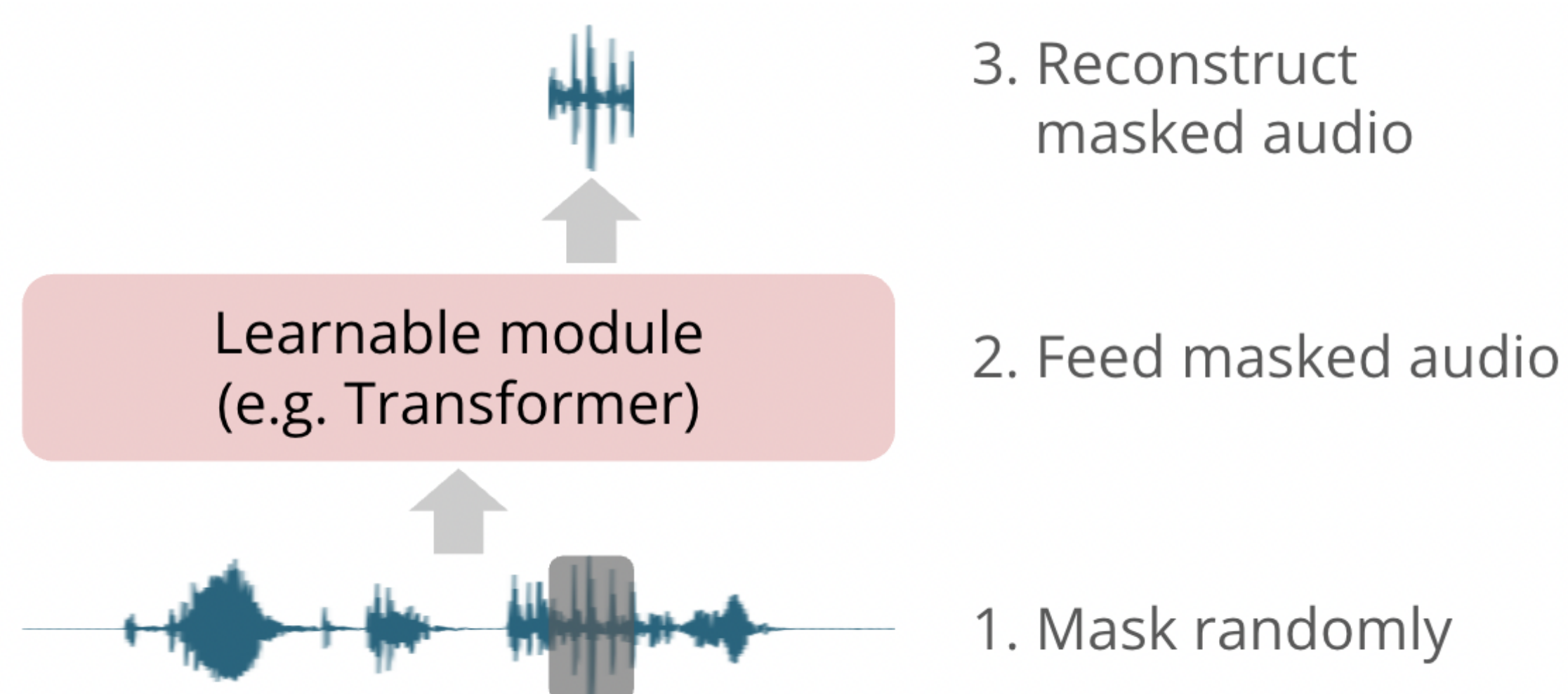


Figure 1. Training a self-supervised speech model.

How do we assess atypical speech?

- Goodness of Pronunciation (GoP)** is conventionally used for atypical pronunciation assessment, such as pathological or non-native speech.
- Phoneme-level pronunciation scores can detect mispronounced phonemes.

Experimental Results

	Dysarthric speech			Non-native speech	
	UASpeech	TORG	SSNCE	s.o.762	L2-ARCTIC
GMM-GoP	0.568	0.682	0.547	0.535	0.243
NN-GoP	0.568	0.681	0.547	0.533	0.312
DNN-GoP	0.548	0.641	0.499	0.380	0.196
MaxLogit-GoP	0.558	0.681	0.547	0.536	0.196
kNN	0.606	0.704	0.523	0.443	0.189
oSVM	0.537	0.671	0.358	0.392	0.082
p-oSVM	0.327	0.621	0.234	0.173	0.078
Ours	0.623	0.713	0.553	0.539	0.182

Table 1. Kendall-tau corr. between pronunciation scores and dysfluency/disfluency (absolute value). Bigger is better.

		GMM-GoP	Ours
Traditional features	MFCC	0.428	0.182
	MelSpec	0.209	0.039
Self-supervised features	XLS-R	0.552	0.602
	WavLM	0.568	0.623

Table 2. Comparing Kendall-tau corr. of GMM-GoP and Ours on UASpeech using various speech features.

- Ours achieve **state-of-the-art** on four out of five datasets.
- Self-supervised features** are generally more effective compared to traditional features.
- Compared to others, ours is successful with self-supervised features, whereas it is significantly worse for traditional features.

Our method: MixGoP

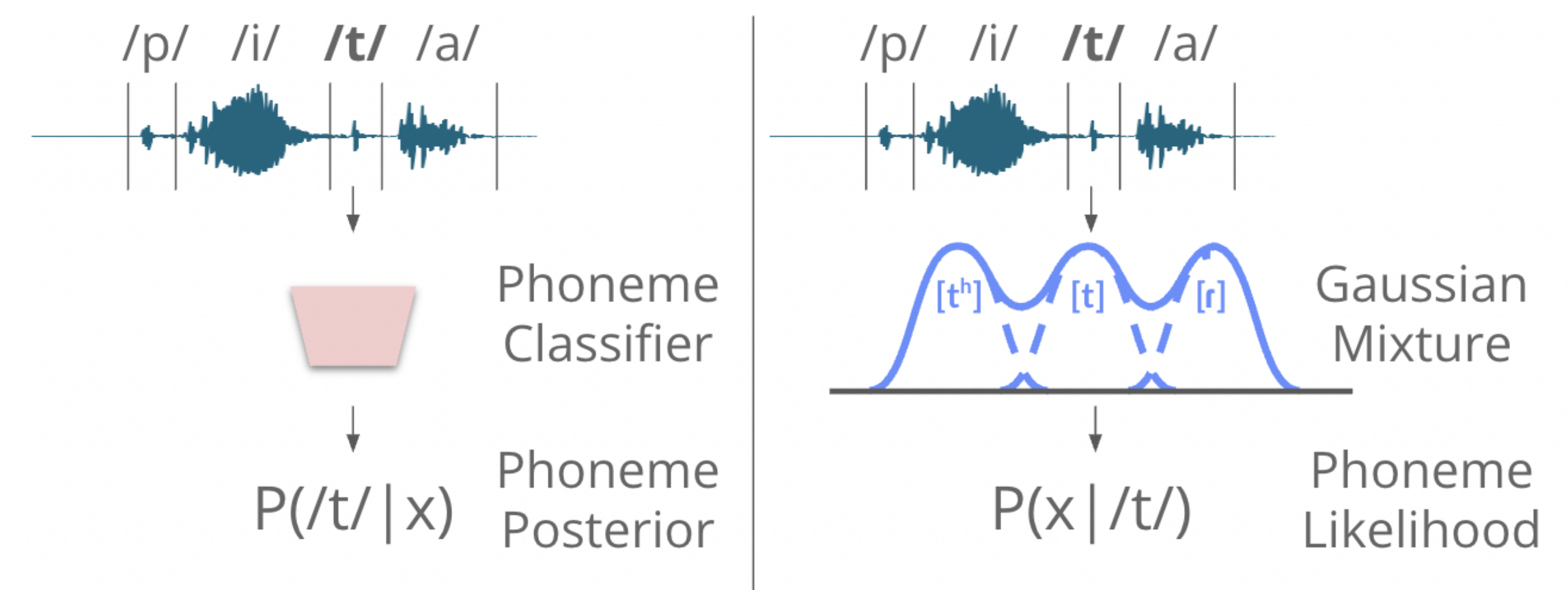


Figure 2. GoP (Previous work) vs. MixGoP (Ours)

Limitations of GoP

- Using fully connected layer assumes **unimodality**. However, single phoneme has **multiple allophones**.
- Using softmax assumes **in-distribution**. However, atypical speech is **out-of-distribution**.

MixGoP: Mixture distribution-based GoP

- Gaussian mixtures** can model allophones.
- Softmax (phoneme posterior) is replaced with **phoneme likelihood** to model out-of-distribution better.

Allophony of S3M features

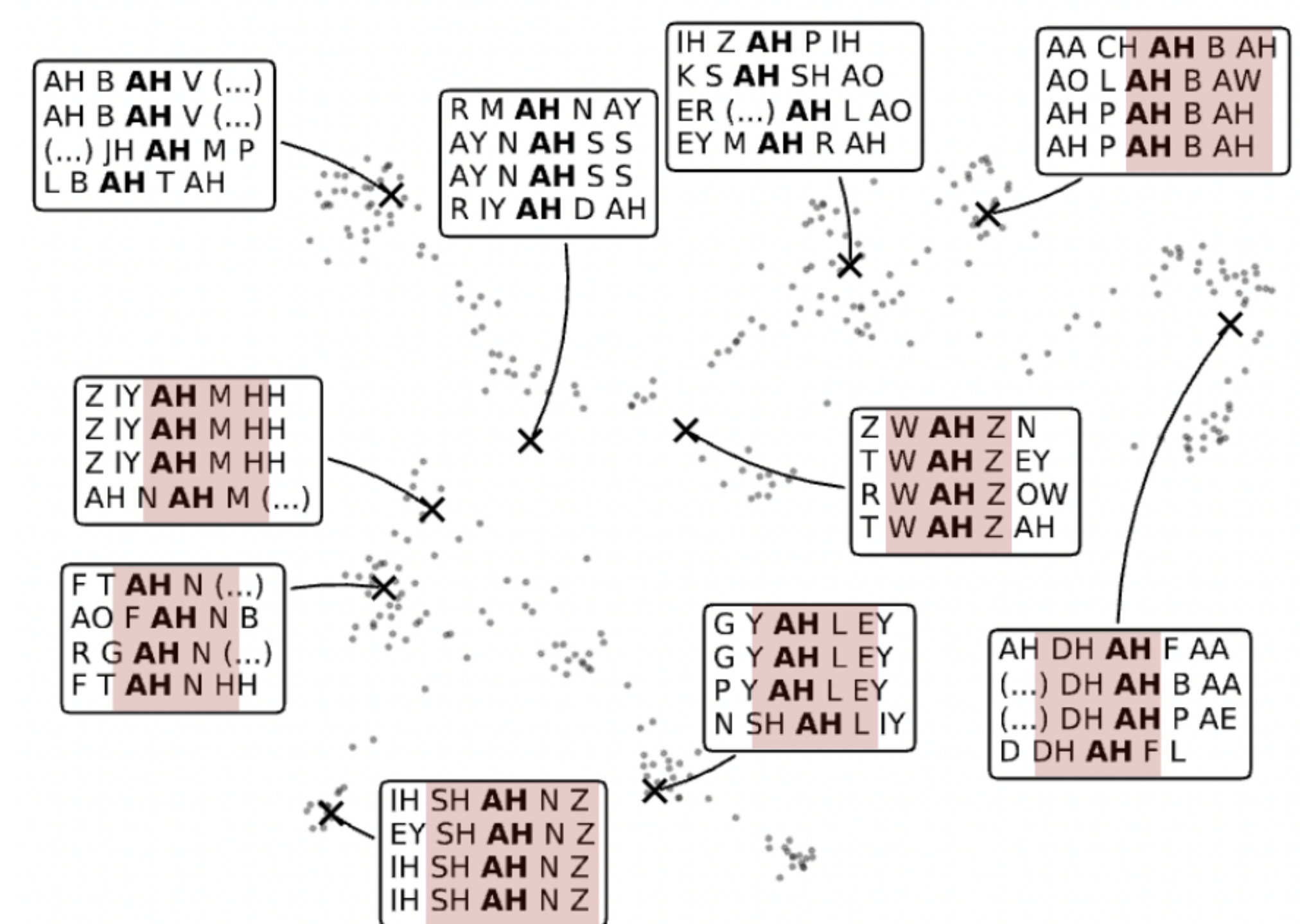
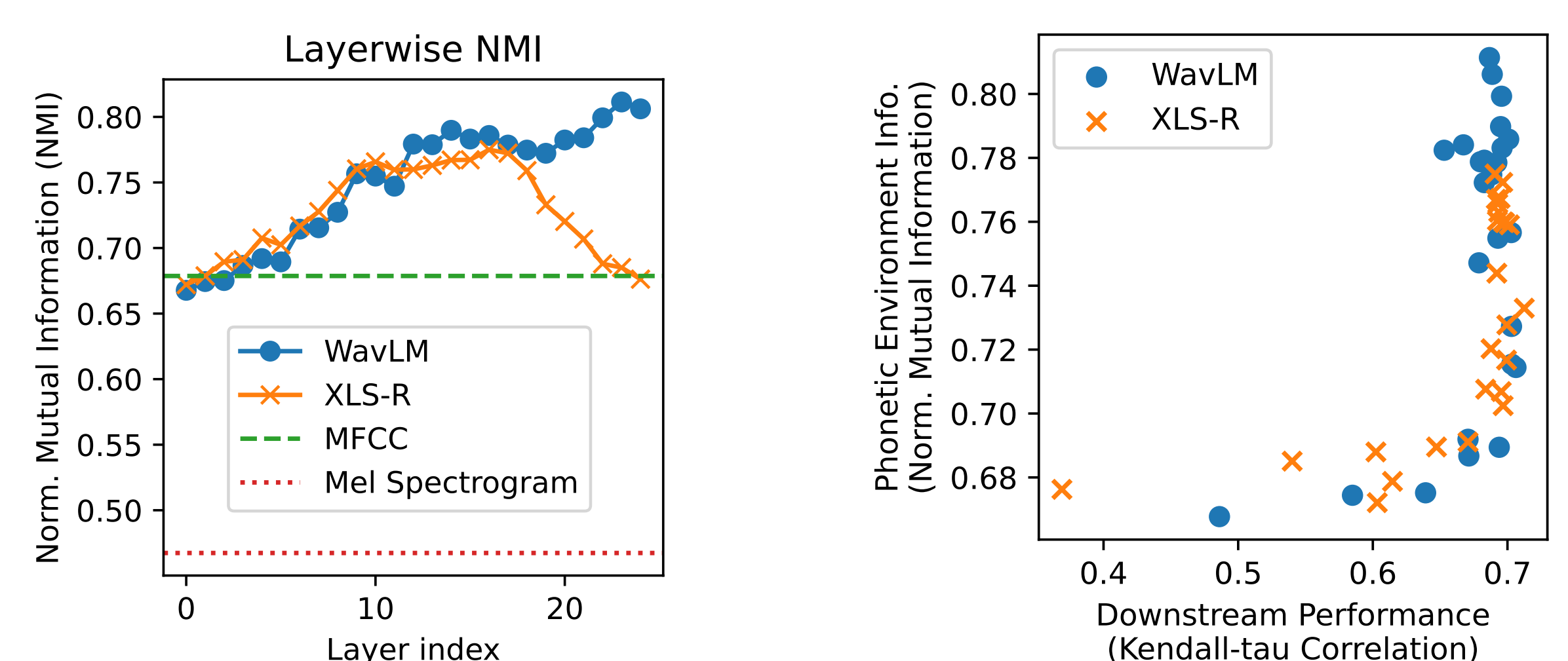


Figure 3. Visualization of WavLM-Large features of the /AH/ phoneme in the TORG healthy subset.



(a) Measuring phonetic environment information on various features

(b) Comparing phonetic environment information with downstream performances

Figure 4. Norm. Mutual Info. between the k-means cluster indices and the phonetic environment on TORG (healthy).

- Self-supervised features are **clustered** based on their **surrounding phonetic environments**.
- Self-supervised features contain more phonetic environment information compared to traditional features.
- Amount of phonetic environment information positively correlates to downstream performance.